SATASHKENT
College Prep Community

COLLEGE PREP COMMUNITY

SAT Math

Problem Solving & Data Analysis

PERCENTAGES, RATES, DATA & INFERENCE • DIGITAL SAT

How to use this file

What is inside

Every question sits alone on its own page, laid out the way it appears in Bluebook: the review bar at the top, the question in the College Board serif, then the four choices or a blank response box.

144 questions in one continuous run across the whole domain: percent change and percent composition, ratios, rates, proportional relationships and unit conversion, mixtures and weighted averages, probability and conditional probability from two-way tables, one-variable data and its displays, scatterplots and lines of best fit, and inference — sampling, study design and margin of error. Topics are interleaved rather than blocked, so you cannot coast on knowing what is coming.

Working through it

Give yourself a strict minute and a half per page. Write the answer down before you turn over. Do not check the key question by question — that costs you the diagnostic.

The answer key

The key is at the back, numbered straight through from 1 to 144. Multiple choice answers are given as a letter. Student produced responses are given as the value you would type into the box.

Where a question allows an equivalent form, the key gives the cleanest one. A fraction and its decimal are both accepted on test day.

Marking

Mark in blocks of twenty rather than all at once. Anything under 80 percent across a block means that topic is worth a second pass before you move to timed sets.

1 Mark for Review

A laboratory analyzed a batch of animal feed and determined that the batch contained 96 kilograms of soybean meal, which made up approximately 32% of the batch's mass. Of the remaining mass of this batch, approximately 57.5% was corn and the remainder was bran. Based on these approximations, to the nearest tenth, how many kilograms of the batch's mass was corn?

ANSWER

2 Mark for Review

Votes by ward, Harrow County

Candidate	Ward 1	Ward 2	Ward 3	Total
Candidate P	70,000	60,000	40,000	170,000
Candidate Q	65,000	58,000	42,000	165,000
Other candidates	15,000	14,000	8,000	37,000
Total	150,000	132,000	90,000	372,000

The table shows the number of votes each candidate received in each of the three wards of Harrow County in a recent election. The three wards together have 806,000 residents and cover a combined area of 1,240 square miles. For these three wards, which of the following is closest to the number of residents who voted per square mile?

A 30

B 137

C 300

D 650

3 Mark for Review

Bike-share trips by bicycle type and purpose

Bicycle type	Commute	Leisure	Total
Standard bicycle	90	150	240
Electric bicycle	120	40	160
Total	210	190	400

The table shows the 400 trips taken on one day using a town's bike-share scheme, classified by the type of bicycle used and the purpose of the trip. If one of these 400 trips is selected at random, what is the probability that the selected trip was taken on an electric bicycle and was taken for leisure?

A $\frac{1}{10}$

B $\frac{1}{4}$

C $\frac{3}{10}$

D $\frac{2}{5}$

4 Mark for Review

Apprentices Completing Certification, by Year

Training center	2016	2017	2018	2019	2020
North Ridge	92	85	78	74	71
Harbor	48	57	66	74	80
Total	140	142	144	148	151

The table shows the number of apprentices who completed a certification program at two training centers in each of five years. No apprentice attended both centers. Which of the following statements are true about the numbers of apprentices who completed the program for the 5 years shown?

I. The mean number from North Ridge Center is greater than the mean number from Harbor Center.

II. The median number from North Ridge Center is greater than the median number from Harbor Center.

A I only

B II only

C I and II

D Neither I nor II

5 Mark for Review

A plant nursery stocks roses, tulips, ferns, cacti, and orchids. Of these plants, 35% are roses, 30% are tulips, 15% are ferns, 12% are cacti, and 24 of the plants are orchids. If the nursery stocks no other plants, how many plants does it stock?

A 200

B 240

C 300

D 480

6 Mark for Review

A certain reed plant grows 6 centimeters every d days. At this rate, which expression represents the number of centimeters, on average, the reed plant grows every w weeks? (1 week = 7 days)

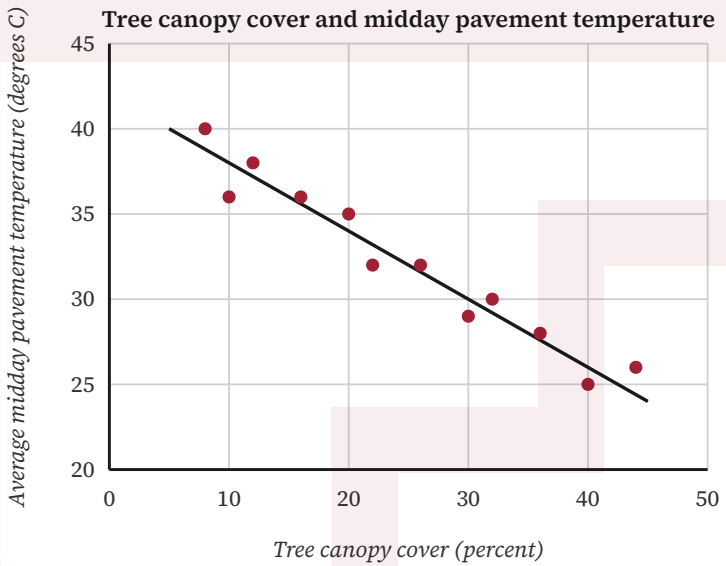
A $\frac{6w}{7d}$

B $\frac{6d}{7w}$

C $\frac{42d}{w}$

D $\frac{42w}{d}$

7 Mark for Review



For each of 12 neighborhoods in a city, an urban planner recorded the percent of the neighborhood's area covered by tree canopy, x, and the average midday pavement temperature, y, in degrees Celsius, on one summer day. The scatterplot shows these data along with the line of best fit, whose equation is $y = 42 - 0.4x$. Which of the following is the best interpretation of the number -0.4 in this equation?

A. The predicted average midday pavement temperature decreases by 42 degrees Celsius for each increase of 1 percentage point in tree canopy cover.

B. The predicted average midday pavement temperature increases by 0.4 degrees Celsius for each increase of 1 percentage point in tree canopy cover.

C. The predicted average midday pavement temperature is 0.4 degrees Celsius for a neighborhood with no tree canopy cover.

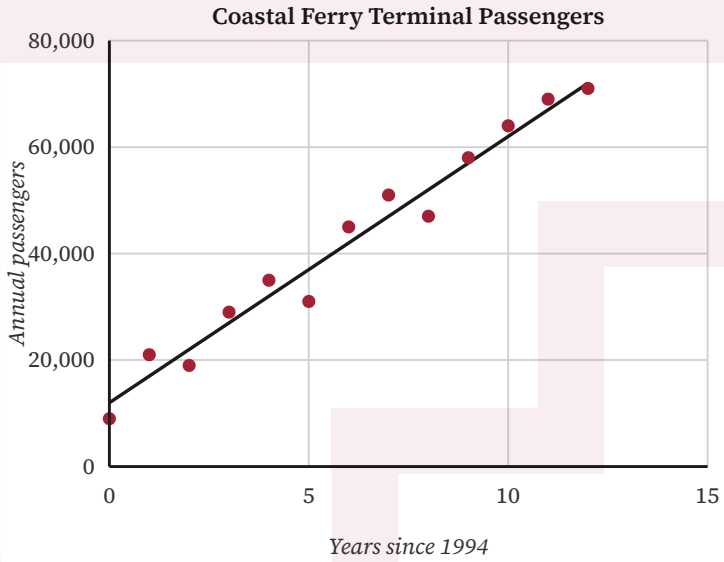
D. The predicted average midday pavement temperature decreases by 0.4 degrees Celsius for each increase of 1 percentage point in tree canopy cover.

8 Mark for Review

The positive number a is 640% of the sum of the positive numbers b and c, and b is 25% of c. What percent of b is a?

ANSWER

9 Mark for Review



The scatterplot shows the number of passengers boarding at a coastal ferry terminal each year from 1994 to 2006, where t is the number of years since 1994 and n is the number of passengers. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

A $n = 5{,}000 + 12{,}000t$

B $n = 9{,}000 + 12{,}000t$

C $n = 12{,}000 + 5{,}000t$

D $n = 12{,}000 + 9{,}000t$

10 Mark for Review

A market vendor can buy 6 bundles of dried lavender for 15 dollars, and she can sell 4 bundles of dried lavender for 26 dollars. How many bundles of dried lavender would she have to buy and then sell to make a total profit of 240 dollars?

ANSWER

11 Mark for Review

The positive number a is 1,500% of the sum of the positive numbers b and c. The number b is 60% of c, and c is 250% of another positive number d. What percent of d is a?

ANSWER

12 Mark for Review

In order to determine whether Compound R is successful in reducing joint stiffness, a research study was conducted. A total of 900 participants were selected at random from a large population of adults who experience joint stiffness. Half of the participants were randomly assigned to receive a placebo treatment (control) and the other half received Compound R. The results indicated that most participants who received Compound R experienced significantly worse joint stiffness as compared to the placebo treatment (perhaps due to a side effect). Based on the design and results of the study, which of the following conclusions is most appropriate?

A Compound R is likely to reduce joint stiffness in adults who do not experience joint stiffness.

B Anyone who receives Compound R will experience worse joint stiffness.

C Compound R is likely to worsen joint stiffness in adults who experience joint stiffness.

D Compound R is likely to worsen joint stiffness in all adults.

13 Mark for Review

On a passenger train carriage, 342 of the 360 seats were sold. What percent of the seats were NOT sold?

A 0.5%

B 5%

C 18%

D 20%

14 Mark for Review

Data set M: 4, 7, 7, 10

Data set N: 4, 7, 7, 10, 32

The lists give the values in data sets M and N. Which statement correctly compares the mean of data set M and the mean of data set N?

A The mean of data set M is greater than the mean of data set N.

B The mean of data set M is less than the mean of data set N.

C The means of data set M and data set N are equal.

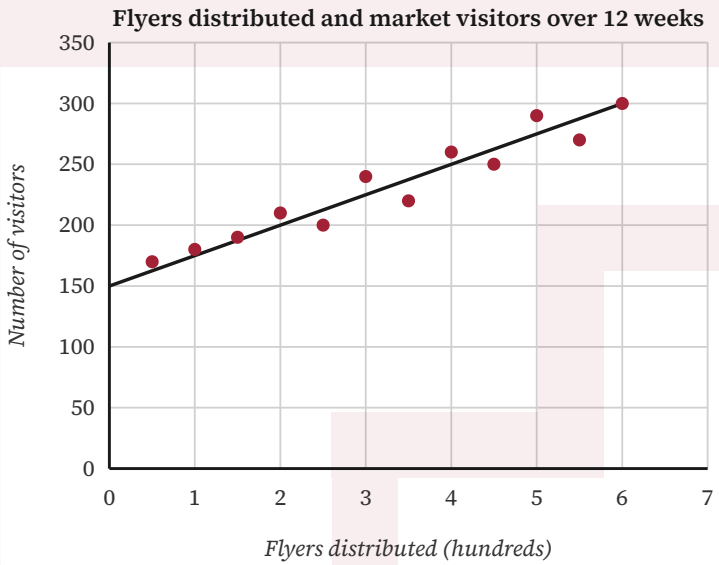
D There is not enough information to compare the means.

15 Mark for Review

An engine coolant is 85% glycol and 15% water. If half the water evaporates, what percent of the new mixture is glycol? (Round to the nearest tenth.)

ANSWER

16 Mark for Review



Each week for 12 weeks, the organizer of a craft market recorded the number of flyers distributed that week, x, in hundreds, and the number of visitors to the market that week, y. The scatterplot shows these data along with the line of best fit. Which of the following is the best interpretation of the y-intercept of the line of best fit?

A In a week when no flyers are distributed, the number of visitors is predicted to be 150.

B In a week when no flyers are distributed, the number of visitors is predicted to be 25.

C For each additional 100 flyers distributed, the predicted number of visitors increases by 150.

D The greatest number of visitors predicted by the line of best fit for the weeks shown is 150.

17 Mark for Review

A solid sphere has a radius of 15 cm. If the radius is decreased by 20%, by what percent does the volume of the sphere decrease?

ANSWER

18 Mark for Review

A shipping depot processes a total of 350 parcels in one day. Each parcel is routed to exactly one of three destinations: the north hub, the east hub, or the south hub. If one of these parcels is selected at random, the probability of selecting a parcel routed to the north hub is 0.38 and the probability of selecting a parcel routed to the east hub is 0.26. How many parcels are routed to the south hub?

ANSWER

19 Mark for Review

Each 2.5-ounce serving of pumpkin seeds and each 4.0-ounce serving of black beans provides about 1 milligram of iron. If a total of 2.3 milligrams of iron are obtained from eating x ounces of pumpkin seeds and y ounces of black beans, which equation best represents this situation?

A $0.25x + 0.4y = 2.3$

B $0.4x + 0.25y = 2.3$

C $2.5x + 4.0y = 2.3$

D $4.0x + 2.5y = 2.3$

20 Mark for Review

A logistics company recorded the number of parcels it delivered to the northern, central, and southern regions. The number of parcels delivered to the central region was 62.5% of the number delivered to the southern region, and the number delivered to the northern region was 40% of the number delivered to the central region. If 5,000 parcels were delivered to the northern region, what was the sum of the numbers of parcels delivered to the central and southern regions?

ANSWER

21 Mark for Review

A college wants to determine the commuting preferences of the students in its evening programme. Which of the following survey methods is most likely to provide the most valid results?

A. Selecting a random sample of 700 students from the college

B. Selecting a random sample of 350 students from the college's evening programme

C. Selecting a random sample of 700 students from the college's evening programme

D. Selecting a random sample of 700 students who park in the college's main car park

22 Mark for Review

A water utility reported that households used 288 million liters of water last year, which was 60% of all the water the utility supplied. Of the remaining water supplied, 75% went to industry and the rest went to municipal services. How many million liters of water went to industry?

ANSWER

23 Mark for Review

A square map has a side length of 60 inches, and 1 inch on the map represents an actual distance of 24 miles. A smaller version of the same map is printed as a square with a side length 60% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A 9.60

B 14.40

C 40.00

D 60.00

24 Mark for Review

A researcher recruited 240 adults who volunteered in response to a poster displayed at a single community center. Each of the 240 volunteers was randomly assigned to one of two groups: one group completed a 10-minute stretching routine each day for eight weeks, and the other group did no stretching routine. At the end of the eight weeks, the volunteers who completed the stretching routine reported significantly fewer episodes of lower back pain than the volunteers who did not. Which of the following is the most appropriate conclusion from this study?

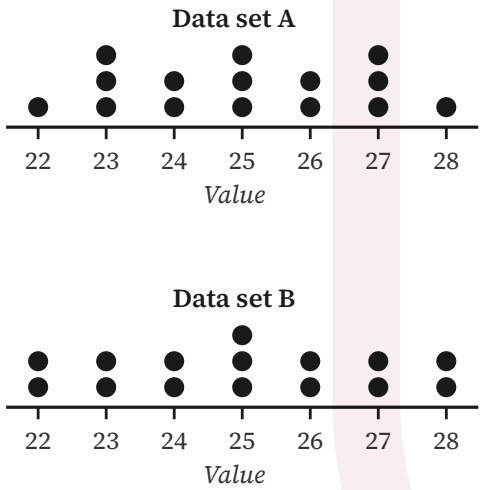
A. The stretching routine caused a reduction in episodes of lower back pain for the volunteers in this study.

B. The stretching routine causes a reduction in episodes of lower back pain for all adults.

C. The stretching routine causes a reduction in episodes of lower back pain for all adults who use community centers.

D. No cause-and-effect conclusion is possible, because the volunteers were not selected at random from a larger population.

25 Mark for Review



The dot plots represent the distributions of values in data sets A and B. Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

26 Mark for Review

A 600-kilogram batch of trail mix is 45% oats, 25% dried fruit, and the rest nuts. A supplier then reduces the dried fruit in the batch by 20%. How many kilograms of dried fruit remain?

ANSWER

27 Mark for Review

Protein and cost by food

Food	Protein	Cost
1 cup of lentils	16 grams	$0.48
1 serving of tofu	10 grams	$0.40

The table shows the amount of protein in two foods and the cost of each food. Based on the table, what is the ratio of the cost per gram of protein in a cup of lentils to the cost per gram of protein in a serving of tofu?

A 1 : 2

B 2 : 3

C 3 : 4

D 4 : 3

28 Mark for Review

A machine begins a shift with a full spool of cable and unwinds cable at a constant rate throughout the shift. What type of model is best to model the length of cable remaining on the spool at any given time during the shift?

A) A linear model with a positive slope

B) A linear model with a negative slope

C) An exponential growth model

D) An exponential decay model

29 Mark for Review

In a hotel, 30% of the rooms are suites and 70% are standard rooms. Balconies are attached to 45% of the suites and 15% of the standard rooms. What percent of all the rooms in the hotel have balconies?

ANSWER

30 Mark for Review

An urban planner conducted research to determine whether there is a relationship between the number of dedicated bicycle lanes per 10,000 residents and the average commute time in a town. Data were collected from a random sample of 120 European towns, and the planner found significant evidence that the average commute time is shorter in towns with more bicycle lanes per 10,000 residents. Which of the following conclusions is best supported by these results?

A. In European towns, there is a positive association between the number of bicycle lanes per 10,000 residents and the average commute time.

B. In European towns, there is a negative association between the number of bicycle lanes per 10,000 residents and the average commute time.

C. In European towns, a decrease in average commute time is caused by an increase in the number of bicycle lanes per 10,000 residents.

D. In European towns, an increase in the number of bicycle lanes per 10,000 residents is caused by a decrease in average commute time.

31 Mark for Review

A publisher's catalog is 40% fiction titles and 60% nonfiction titles. Translated works make up 25% of the fiction titles and 15% of the nonfiction titles. What percent of the catalog is translated works?

ANSWER

32 Mark for Review

A cannery purchases pickling brine in 24-gallon barrels. The brine is mixed with water to make several batches of solution. Each batch of solution is made by mixing 6 pints of brine with water. What is the maximum number of batches of solution that can be made from one 24-gallon barrel of brine? (1 gallon = 8 pints)

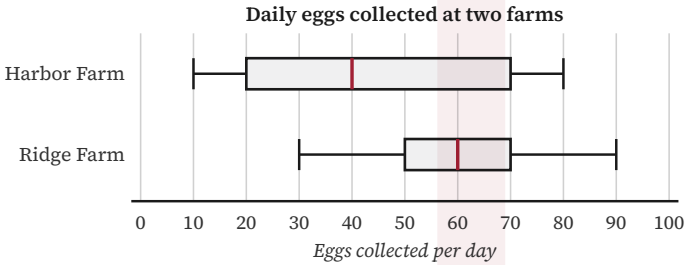
A 4

B 24

C 32

D 192

33 Mark for Review



The box plots summarize the number of eggs collected each day at two poultry farms, Harbor Farm and Ridge Farm, on each of 45 days. The interquartile range of the daily numbers for Harbor Farm is how much greater than the interquartile range of the daily numbers for Ridge Farm?

A 10

B 20

C 30

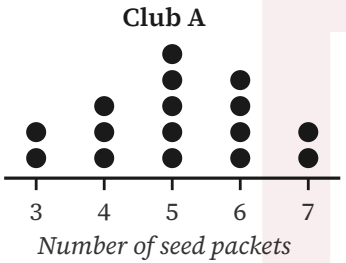
D 50

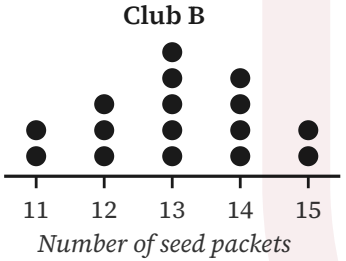
34 Mark for Review

A rectangular solar panel array has a length of 30 m and a width of 18 m. If the length is increased by 40% and the width is increased by 15%, by what percent does the area of the array increase?

ANSWER

35 Mark for Review





Each of the dot plots shown represents the number of seed packets contributed by each member of two garden clubs, club A and club B. Which statement best compares the standard deviations of the numbers of seed packets contributed by each member for these two clubs?

A. The standard deviation of the number of seed packets contributed by each member for club A is less than the standard deviation of the number of seed packets contributed by each member for club B.

B. The standard deviation of the number of seed packets contributed by each member for club A is equal to the standard deviation of the number of seed packets contributed by each member for club B.

C. The standard deviation of the number of seed packets contributed by each member for club A is greater than the standard deviation of the number of seed packets contributed by each member for club B.

D. There is not enough information to compare these standard deviations.

36 Mark for Review

The density of an object is equal to its mass divided by its volume, and silver has a density of 10.5 grams per cubic centimeter. A piece of alloy is 40% silver by volume. If the volume of the piece of alloy is 15 cubic centimeters, what is the mass of the silver, in grams, in the piece of alloy?

A 0.28

B 6.00

C 63.00

D 157.50

37 Mark for Review

A cylindrical grain silo has a radius of 3 m and a height of 12 m. If the radius is increased by 20% and the height is increased by 25%, by what percent does the volume increase?

ANSWER

38 Mark for Review



The graph shows the number of electric scooters serviced each month for six months at each of two service depots. Based on the graph, what was the average rate of change, in scooters per month, in the number of scooters serviced at the Meadow depot from month 1 to month 5?

A 10

B 15

C 16

D 20

39 Mark for Review

The density of dry sand is 1,600 kilograms per cubic meter. What is the density, in pounds per cubic foot? (Round to the nearest whole number; 1 kilogram = 2.205 pounds, 1 cubic foot = 0.0283 cubic meters)

ANSWER

40 Mark for Review

In a warehouse, 35% of the total floor area is cold storage and the remaining area is dry storage. Shelving covers exactly 24% of the cold storage area and exactly 16% of the dry storage area. If shelving covers exactly $q\%$ of the total floor area of the warehouse, what is the value of q?

ANSWER

41 Mark for Review

Researchers must conduct an experiment to see whether a new mouth rinse is effective in reducing dental plaque. They have selected a random sample of 120 dental patients. Some of the patients are assigned to the new mouth rinse while the rest are assigned to the standard mouth rinse. Which of the following methods of assigning each patient's treatment is most likely to lead to a reliable conclusion about the effectiveness of the new mouth rinse?

A Patients over the age of 40 are assigned the new mouth rinse.

B Those who already have more than one dental filling are assigned the new mouth rinse.

C The patients divide themselves evenly into two groups. A coin is tossed to decide which group receives the new mouth rinse.

D Each patient is assigned a random number. Those with an odd number are assigned the new mouth rinse.

42 Mark for Review

A

Value	Frequency
40	7
50	6
60	6
70	7

B

Value	Frequency
40	9
50	4
60	4
70	9

C

Value	Frequency
40	5
50	6
60	7
70	8

D

Value	Frequency
40	6
50	6
60	6
70	6

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

(A) A

(B) B

(C) C

(D) D

43 Mark for Review

A quality inspector approved 85% of the production batches checked in one week. If 6 batches were rejected, how many batches were approved?

A 24

B 30

C 34

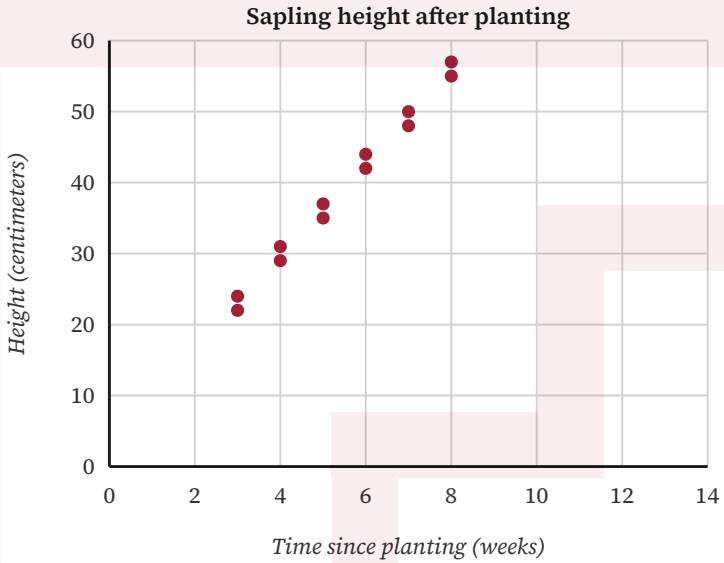
D 40

44 Mark for Review

A dispenser holds a drink that is 60% orange juice and 40% sparkling water. If 20% of the orange juice is poured out and none of the sparkling water is removed, what percent of the remaining drink is orange juice? (Round to the nearest tenth.)

ANSWER

45 Mark for Review



The scatterplot shows data set A, which consists of the heights y, in centimeters, of a sapling at various times x, in weeks, after planting. The equation of a line of best fit for the relationship in data set A can be written as $y = 4.0 + 6.5x$, where $3 \leq x \leq 8$. The sapling was measured again 14 weeks after planting and was 62 centimeters tall. Data set B consists of all the data points in data set A as well as the data point (14, 62). The equation of a line of best fit for data set B can be written as $y = r + sx$, where r and s are constants and $3 \leq x \leq 14$. Assuming the equations of the lines of best fit are calculated in the same way, which of the following is the best estimate for the value of s?

A 4.0

B 6.5

C 9.1

D 11.7

46 Mark for Review

At an airport, 65% of the land is runways and 35% is terminals. Solar panels cover 8% of the runway land and 24% of the terminal land. What percent of the total land is covered by solar panels?

ANSWER

47 Mark for Review

For a study, a group of apple trees will be selected from an orchard of 290 apple trees, and a group of pear trees will be selected from an orchard of 350 pear trees. Some of the apple trees and pear trees will receive a new pruning method, and some of the apple trees and pear trees will receive the orchard's standard pruning method. Which of the following is necessary for this study to attempt to establish a cause-and-effect relationship between two variables?

A. The total number of apple trees in the study is equal to the total number of pear trees in the study.

B. The apple trees and pear trees are randomly selected from their orchards.

C. The proportion of young to mature pear trees receiving the new pruning method is equal to the proportion of young to mature apple trees receiving the new pruning method.

D. The apple trees and pear trees are randomly assigned to the new pruning method and the standard pruning method.

48 Mark for Review

A drum of window-cleaning fluid is 50% cleaning concentrate and 50% water. If 75% of the water evaporates, what percent of the remaining fluid is cleaning concentrate?

ANSWER

49 Mark for Review

A magazine's number of subscribers first increased by 30% and then the new number decreased by 20%. The final number of subscribers is what percent of the beginning number?

A 96%

B 104%

C 110%

D 114%

50 Mark for Review

A regional library service is proposing a change to the Sunday opening hours at all of its branches. A sample of 418 library cardholders was selected at random from all library cardholders in the region. The selected cardholders were asked whether they approved of the proposed change, and 355 cardholders responded that they did not approve. Which of the following is the largest population that the results of the survey can be generalised to?

A All library cardholders in the region

B The 418 cardholders who were surveyed

C The 355 cardholders who responded that they did not approve of the proposed change

D All library cardholders in the country

51 Mark for Review

On March 1, a mobile network region had 54 million active lines. By August 1 of the same year, the number of active lines in the region had increased to 76 million. If an active line in the region on August 1 is selected at random, the probability that the line is served by the Kestrel network is 0.35. There were p million Kestrel lines in the region on August 1. To the nearest integer, what is the value of p?

ANSWER

52 Mark for Review

At a repair shop, 50% of the devices brought in are phones and 50% are tablets. Screen replacements are needed by 34% of the phones and 22% of the tablets. What percent of all the devices brought in need a screen replacement?

ANSWER

53 Mark for Review

A tank has a volume of 0.72 cubic meters. Which of the following is closest to this volume, in gallons? (Note: 100.00 centimeters = 1 meter. Use 3,785.41 cubic centimeters = 1.00 gallon.)

A 0.19

B 1.90

C 19.02

D 190.20

54 Mark for Review

Hours of Volunteer Service

14	22	9
18	31	12
25	16	11
27	19	13
96	15	21
88	17	23

A newsletter reported the mean number of hours of service completed by the volunteers in a community project. The hours completed by the 18 volunteers in the project are given in the table shown. Which of the following statements best describes why the mean number of hours alone may not be a useful measure of centre for the distribution?

(A) Outliers may cause the distribution to be skewed.

(B) Repeated numbers may cause the distribution to be skewed.

(C) The number of volunteers is too small to calculate and report the mean.

(D) Sets containing an even number of data points may cause the distribution to be skewed.

55 Mark for Review

A sample of ore is 62% iron oxide, 26% silica, and the rest alumina. If the mass of alumina in the sample was found to be 45.6 kg, how many kilograms of iron oxide does the sample contain?

ANSWER

56 Mark for Review

Sample	Percent in favour	Margin of error
A	63.41%	8.2%
B	70.28%	4.9%

The results of two random samples of ferry passengers asked about a proposed new timetable are given in the table shown. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

A Sample A had a larger sample size.

B Sample A had a smaller sample size.

C Sample A had a smaller number of responses that could not be recorded.

D Sample A had a lower percent of passengers in favour of the proposed new timetable.

57 Mark for Review

At a bottling plant, 0.02% of all bottle caps produced are defective. On average, there will be three defective caps out of how many produced?

A 1,500

B 6,000

C 15,000

D 150,000

58 Mark for Review

A charity received a grant of 15,000 dollars, which made up 20% of its income for the year. Of the rest of its income, 45% came from individual donations and the remainder came from fundraising events. How many dollars of the charity's income came from individual donations?

ANSWER

59 Mark for Review

Number of jars produced

Inspections per month	12 to 19 jars	20 or more jars	Total
4 to 6 times	8	7	15
More than 6 times	5	25	30
Total	13	32	45

A beekeeping association wanted to study the relationship between the number of jars of honey a hive produces in a season and the number of times each hive is inspected per month. For the study, the association selected 70 hives at random from all hives registered with the association. The table displays the information for the 45 hives in the sample that were inspected at least 4 times per month.

Which of the following is the largest population to which the results of the study can be generalised?

A All hives registered with the association

B The 70 hives in the sample

C The 45 hives in the sample that were inspected at least 4 times per month

D All hives registered with the association that were inspected at least 4 times per month

60 Mark for Review

A regional train traveled nonstop between two cities, a total distance of 400 kilometers. The train traveled at a speed of 96 kilometers per hour for the first 45 minutes of the trip and at a speed of 110 kilometers per hour for the remainder of the trip. To the nearest minute, for how many minutes did the train travel at a speed of 110 kilometers per hour?

ANSWER

61 Mark for Review

Pass type and attendance at the preview night

Pass type	Attended	Did not attend	Total
Standard pass			240
Premium pass	40	80	120
Total			360

A theatre club has 360 members. Each member holds either a standard pass or a premium pass. The table shows some information about whether the members attended the club's preview night, with the values in some of the cells not shown. The probability that a randomly selected member of the club holds a standard pass and attended the preview night is $\frac{1}{3}$. If a member who attended the preview night is selected at random, what is the probability that the selected member holds a premium pass?

A $\frac{1}{9}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{1}{2}$

62 Mark for Review

Data set J: 2, 4, 6, 8, 10, 12, 14

Data set K: 2, 2, 4, 4, 6, 6, 8

Which of the following statements correctly compares the means of data set J and data set K?

A The mean of each data set is 4.

B The mean of each data set is 8.

C The mean of data set J is less than the mean of data set K.

D The mean of data set J is greater than the mean of data set K.

63 Mark for Review

A partially filled tank containing 42 liters of coolant is connected to a pump that delivers one 0.6-liter pulse of coolant every 4 seconds. Until the tank is full, which of the following can be used to represent the volume v, in liters, of coolant in the tank t seconds after the pump starts, where t is a multiple of 4?

A $v = 4t$

B $v = 0.15t + 42$

C $v = 0.6t + 42$

D $v = 2.4t + 42$

64 Mark for Review

In a city's bus network, 45% of the routes are express routes and 55% are local routes. Night service is offered on 20% of the express routes and 8% of the local routes. What percent of all the routes in the network offer night service?

ANSWER

65 Mark for Review

A city transit authority surveys rider satisfaction by texting a questionnaire to every rider registered with a smartcard account. Unknown to the authority, 35% of riders pay their fares in cash and therefore have no smartcard account. If 55% of smartcard riders are satisfied but only 30% of cash-paying riders are, what is the likely effect on the survey's estimate?

A Overestimates satisfaction due to undercoverage of cash-paying riders

B Underestimates satisfaction due to nonresponse bias

C Accurately reflects satisfaction since every account holder was contacted

D Underestimates satisfaction due to undercoverage of cash-paying riders

66 Mark for Review

A cart's speed is increasing at a rate of 8.40 feet per second squared. What is this rate in kilometers per minute squared, rounded to the nearest tenth? (Use 1 kilometer = 3,281 feet.)

ANSWER

67 Mark for Review

A conservation group recorded the number of nest boxes occupied at each of 9 monitoring sites. The 9 values, listed in increasing order, are 10, 12, 14, 15, 19, 19, 19, 20, and 70. The group then discovered that the value 70 belonged to a different survey and removed it from the data set, leaving 8 values. Removing this value decreased the mean of the data set by m and decreased the median of the data set by n. What is the value of $m - n$?

ANSWER

68 Mark for Review

Year of training	1	2	3
Number of apprentices	24	33	27

A company conducted a study concerning the number of hours per week spent on paid part-time work by apprentices in years 1, 2 and 3 of their training and their self-reported stress level. A total of 84 apprentices were selected at random from all apprentices in years 1, 2 and 3 registered with one large training provider in Canada. The selected apprentices were asked to report their stress level as either low, medium, or high. The table summarises the distribution of the selected apprentices by year of training.

If the results of the study suggest a relationship between the self-reported stress level and the number of hours per week spent on paid part-time work by apprentices in years 1, 2 and 3, which of the following aspects of the study will prevent the results from being generalised to all apprentices in years 1, 2 and 3 in Canada?

A. The selected apprentices do not all work for the same employer.

B. The number of selected apprentices is too small.

C. The number of selected apprentices from each year of training is different.

D. The apprentices were not selected at random from all apprentices in years 1, 2 and 3 in Canada.

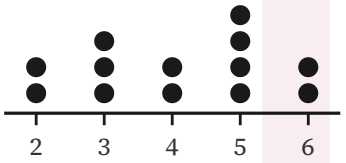
69 Mark for Review

By 9:00 AM, a total of 30 sacks of grain had been loaded onto a barge. By 11:30 AM, a total of 60 sacks of grain had been loaded onto the same barge. If sacks are loaded onto the barge at a constant rate, what is the total number of sacks that will have been loaded onto the barge by 4:00 PM?

ANSWER

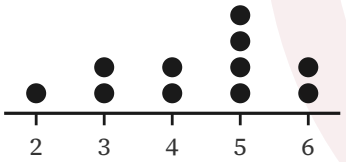
70 Mark for Review

Elm Street site (13 mornings)



Number of cars waiting at opening

Vale Road site (11 mornings)



Number of cars waiting at opening

The dot plots show the number of cars waiting at a car wash at the moment it opened, recorded on each of 13 mornings at the Elm Street site and on each of 11 mornings at the Vale Road site. What is the median of the combined data set consisting of all 24 recorded values?

ANSWER

71 Mark for Review

Annual Rail Passengers, in millions

Rail line	2019	2022
Northgate	78.4	81.2
Riverbend	62.9	66.5
Ashmore	53.6	51.9
Tenby	41.6	44.0
Loxley	33.2	36.7
Ferncross	24.8	29.6
Selkirk	22.3	26.4

The table shows the number of annual rail passengers, rounded to the nearest tenth of a million, carried by each of the seven busiest regional rail lines in a country in 2019 and in 2022. Based on the information given in the table, how much greater, in millions, was the median number of annual rail passengers on these seven lines in 2022 than the median number in 2019, to the nearest tenth of a million?

ANSWER

72 Mark for Review

A number x is 250% greater than a positive number y. A number z is 60% less than the number x. The number z is how many times the number y?

A 0.40

B 1.40

C 2.10

D 3.50

73 Mark for Review

The area covered by a spreading patch of algae in a tank is increasing at a rate of 720 square centimeters per minute. Which of the following is closest to this rate in square inches per second? (Use 1 inch = 2.54 centimeters.)

A 1.86

B 4.72

C 30.48

D 111.60

74 Mark for Review

A transport researcher selected 34 electric delivery bicycles at random from the fleet operating in one city and 22 electric delivery bicycles at random from the fleet operating in a second city. Each city's delivery district was divided into equally sized map sectors, and each bicycle was tracked for a fixed number of hours per shift.

To observe usage patterns, the researcher counted the number of sectors each bicycle entered during its first month of service and again during its sixth month of service. The researcher found that in the first month, the number of sectors entered by the bicycles from the second city's fleet was significantly greater than the number entered by the bicycles from the first city's fleet. The researcher also found that the number of sectors entered by bicycles from both fleets decreased from the first month to the sixth month.

Based on the researcher's findings, which of the following statements is an appropriate conclusion that can be drawn from this study?

A. The statistically significant difference in the number of sectors entered between the two time periods was caused by the differences between the two cities' street layouts.

B. The statistically significant difference in the number of sectors entered between the two time periods was caused by the ageing of the bicycles.

C. The statistically significant difference in the number of sectors entered during the first month between the two groups of bicycles can be generalised to all electric delivery bicycles.

D. The statistically significant difference in the number of sectors entered during the first month between the two groups of bicycles can only be generalised to the two fleets in the study.

75 Mark for Review

On a farm, 70% of the land is cropland and 30% is pasture. Irrigation systems cover 15% of the cropland and 5% of the pasture. What percent of the farm's land is covered by irrigation systems?

ANSWER

76 Mark for Review

A granola recipe mixes oats, seeds, and honey in the ratio 5 : 3 : 2 by mass. A producer wants to make 4.5 kilograms of granola using this recipe. How many kilograms of oats are needed?

A 0.9

B 1.35

C 1.5

D 2.25

77 Mark for Review

A librarian wants to conduct a survey to estimate the mean number of pages read per day by a certain group of people. The librarian has not yet decided which group of people to survey. Surveying a random sample from which of the following groups would most likely result in the smallest margin of error?

A Residents of the same town

B Customers of a certain bookshop

C Subscribers to the same magazine

D Members of a reading group who all follow the same weekly reading schedule

78 Mark for Review

A robotics club has entered 30 contests and has won 50% of the contests it has entered. What is the minimum number of additional contests the club must win in order to finish the season winning 75% of the contests it has entered?

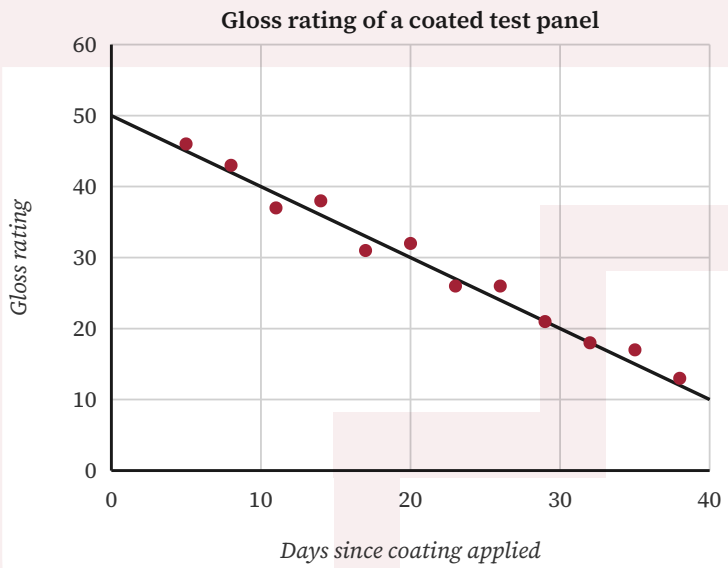
A 15

B 22

C 30

D 45

79 Mark for Review



The scatterplot shows the relationship between the number of days since a protective coating was applied, x, and the gloss rating, y, of a test panel for data set A. A line of best fit for the data is also shown. Data set B is created by subtracting 12 units from the value of y for each data point in data set A. Which of the following is closest to the y-coordinate of the y-intercept of the line of best fit for data set B?

A 26

B 38

C 50

D 62

80 Mark for Review

A vat holds a syrup that is 20% sugar concentrate and the rest water. If 25% of the water evaporates, what is the new percent concentration of sugar concentrate?

ANSWER

81 Mark for Review

In a transit survey of 60 bus riders and 40 train riders, 45% of the bus riders and 70% of the train riders said they commute daily. What percent of the people surveyed said they commute daily?

A 51%

B 55%

C 57.5%

D 62%

82 Mark for Review

A health authority conducted a study on the amount of time spent outdoors each day by adults aged 30 to 50 and their measured vitamin D levels. A total of 4,800 adults were selected at random from all adults aged 30 to 50 living in one large province of a certain country. The selected adults were asked to report their daily time outdoors as less than 30 minutes, 30 minutes to 2 hours, or more than 2 hours. If the results of the study suggest a relationship between the self-reported time spent outdoors and adults' vitamin D levels, which of the following aspects of the study will prevent these results from being generalised to all adults aged 30 to 50 in the country?

A The selected adults do not all live in the same town within the province.

B The number of selected adults is too small.

C The number of selected adults aged 40 to 50 is greater than the number aged 30 to 39.

D The adults were not selected at random from all adults aged 30 to 50 in the country.

83 Mark for Review

Data set R: 12, 12, 12, 12, 12, 12, 12, 12, 12

Data set S: 12, 12, 12, 12, 12, 12, 12, 12, 12, 96

Which of the following statements about the means and medians of data set R and data set S is true?

A Only the means are different.

B Only the medians are different.

C Both the means and the medians are different.

D Neither the means nor the medians are different.

84 Mark for Review

A technician found that a battery pack contained 40 kg of casing, which made up 25% of its total mass. Of the remainder, 75% is cell material and the rest is wiring. How many kilograms of cell material does the battery pack contain?

ANSWER

85 Mark for Review

In 2019, a coastal province had a population of approximately 152,400 people and a population density of 530 people per square kilometer. Based on these approximations, if 14% of the province's area was covered by forest, to the nearest whole number, what is the area, in square kilometers, of forest in the province?

ANSWER

86 Mark for Review

In a survey, 92 employees of a particular company were selected at random, and approximately 35% of them reported that they cycle to work at least once a week. If the reported percentage is used as an estimate for the proportion of all employees at the company who cycle to work at least once a week, the margin of error is 11%. Which of the following statements is appropriate based on the data provided?

A Less than 20% of all the company's employees cycle to work at least once a week.

B Between 0% and 11% of all the company's employees cycle to work at least once a week.

C Between 24% and 46% of all the company's employees cycle to work at least once a week.

D Approximately 11% of the surveyed employees misreported how often they cycle to work.

87 Mark for Review

Three machines share a production run that lasts a total of 4 hours. If the first machine runs for 35% of the total time and the second machine runs for 40% of the total time, for how many minutes does the third machine run?

A 48

B 60

C 72

D 84

88 Mark for Review

Transit app use and bicycle riding

Bicycle riding	Uses transit app daily	Does not use transit app daily	Total
Rare	88	120	208
Occasional	152	96	248
Frequent	131	53	184
Total	371	269	640

In a study of commuting habits, 640 randomly selected adults in a metropolitan area were asked whether they use a transit app daily and how often they ride a bicycle. The data are summarized in the table. If one of the 640 adults surveyed is selected at random, what is the probability that the adult uses a transit app daily?

A $\frac{1}{640}$

B $\frac{269}{640}$

C $\frac{371}{640}$

D $\frac{131}{184}$

89 Mark for Review

The area of a coastal nature reserve is 8 hectares. What is its area, in square yards? (1 hectare = 10,000 square meters, 1 square meter = 1.196 square yards)

ANSWER

90 Mark for Review

An alloy ingot is 55% copper, 30% zinc, and the rest tin. If the ingot contains 12 kg of tin, how many kilograms of copper does it contain?

ANSWER

91 Mark for Review

A study estimates the average amount of food waste produced per week by restaurants in a large city, reporting a 95% confidence interval of 12.6 kg to 15.4 kg based on a random sample of 240 restaurants. Which of the following is the most appropriate conclusion from this research?

A) 95% of restaurants in the city produce between 12.6 kg and 15.4 kg of food waste per week.

B) We are 95% confident that the true mean weekly food waste for restaurants in the city is between 12.6 kg and 15.4 kg.

C) There is a 5% chance that the true mean weekly food waste is exactly 14.0 kg.

D) All restaurants in the sample produced between 12.6 kg and 15.4 kg of food waste per week.

92 Mark for Review

Percent of Homes with Rooftop Solar Panels

Township	Percent of homes
Township A	25.3%
Township B	11.7%
Township C	34.7%
Township D	27.8%
Township E	14.2%
Township F	18.4%
Township G	22.6%

A regional energy agency surveyed all 40 townships in a county and recorded the percent of homes in each township that have rooftop solar panels. The results for 7 of the townships are given in the table. The median percent of homes with rooftop solar panels for all 40 townships was 21.85%. What is the difference between the median percent of homes with rooftop solar panels for these 7 townships and the median for all 40 townships?

A 0.25%

B 0.75%

C 1.35%

D 5.95%

93 Mark for Review

A scrap-metal bale was analyzed. Aluminum made up 64% of its mass, copper made up 25% of the non-aluminum part, and the rest was plastic. If the bale contained 192 kg of aluminum, how many kilograms of copper did it contain?

ANSWER

94 Mark for Review

Growth factors by species

Species of tree	Growth factor
Quaking aspen	2.0
Basswood	3.0
Honey locust	3.5
Bur oak	4.0
Sweetgum	4.5
Sugar maple	5.5
Silver poplar	6.0
Ironwood	7.0

One method of estimating the approximate age, in years, of a tree of a particular species is to multiply the diameter of the tree, in inches, by a constant called the growth factor for that species. The table gives the growth factors for eight species of trees. If a silver poplar and a bur oak each now have a diameter of 2 feet, which of the following is closest to the difference, in inches, of their diameters 10 years from now? (1 foot = 12 inches)

A 0.6

B 0.7

C 0.8

D 0.9

95 Mark for Review

A machine fills sachets with approximately 250 grams of ground coffee. To test the accuracy of the filling process, 186 sachets of coffee were selected at random and weighed. Based on the sample, it is estimated that the average weight of all sachets of coffee filled by the machine during a 12-hour production run is 248.4 grams, with an associated margin of error of 0.35 gram. Which of the following is the best interpretation of this estimate?

A. Plausible values for the average weight of all sachets of coffee filled by the machine are between 248.05 grams and 248.75 grams.

B. Plausible values for the average weight of all sachets of coffee filled by the machine are less than 248.05 grams or greater than 248.75 grams.

C. The average weight of all sachets of coffee filled by the machine is greater than 248.5 grams.

D. The average weight of all sachets of coffee filled by the machine is less than 248.5 grams.

96 Mark for Review

The radius of a circular trampoline is increased by 30%. By what percent does the area of the trampoline increase?

ANSWER

97 Mark for Review

The density of a certain sandstone is 2.4 grams per cubic centimeter. If the volume of a block of this sandstone is 0.25 cubic meters, what is its mass in kilograms?

ANSWER

98 Mark for Review

A polling agency wanted to test whether a proposed membership rule change would pass with greater than 50% yes votes. The agency sampled 1,200 registered members selected at random, and 51.2% of those sampled favoured the rule change. The margin of error associated with this poll was $\pm 4\%$. Based on the poll's results, which of the following statements must be true?

A The percentage of members who will vote yes for the rule change is 51.2%.

B The rule change will pass with more yes votes than no votes, but the percentage of votes it will receive cannot be predicted.

C The rule change will pass with at least 55.2% of the vote.

D The poll's results do not provide sufficient evidence to conclude that the rule change will pass.

99 Mark for Review

The number of daily riders on a shuttle route falls by 20%. By what percent of the new number of riders must the number rise in order to return to its original value?

ANSWER

100 Mark for Review

In a psychological study, researchers asked each participant to complete a timed typing task after taking a rest break, the length of which varied from participant to participant. For each participant the researchers recorded the length of the rest break, x, in minutes, and the number of typing errors made during the task, y. The line of best fit for the data is $y = -0.35x + 24.6$. Which of the following is the best interpretation of the meaning of the y-intercept of the line of best fit?

A. The expected decrease in the number of typing errors made per additional minute of rest break

B. The expected increase in the number of typing errors made per additional minute of rest break

C. The expected length of rest break required for a participant to complete the task with no typing errors

D. The expected number of typing errors made by a participant who takes no rest break

101 Mark for Review

A 200-liter vat holds a mixture that is 30% dye concentrate and the rest water. If 50 liters of water evaporate, what percent of the remaining liquid is dye concentrate?

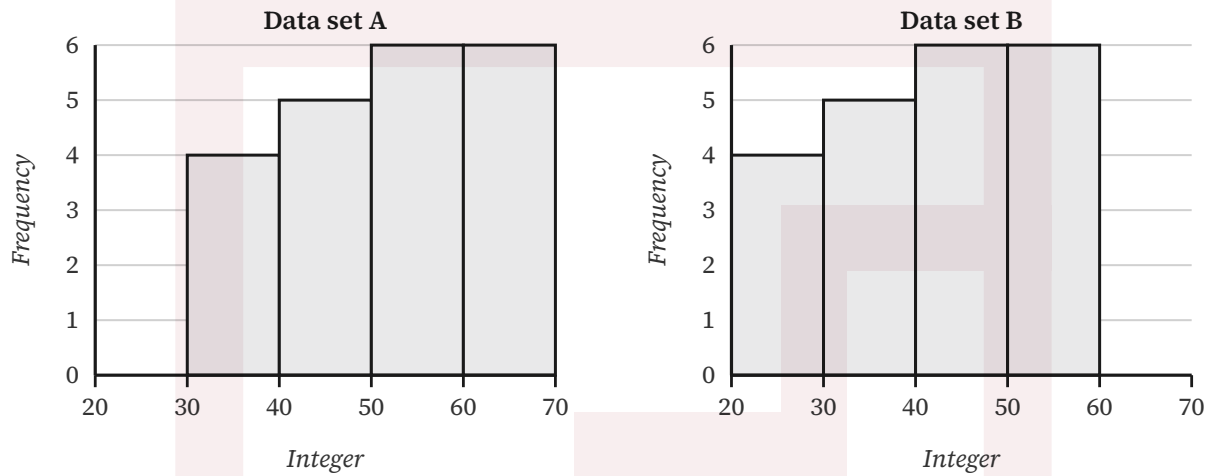
ANSWER

102 Mark for Review

A batch of concrete mix contains 90 kg of cement, which makes up 15% of the batch's mass. Of the remainder, 60% is gravel and the rest is sand. How many kilograms of sand are in the batch?

ANSWER

103 Mark for Review



Two data sets of 21 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 20, but less than 30. The second interval represents the frequency of integers greater than or equal to 30, but less than 40, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

A 0

B 1

C 10

D 21

104 Mark for Review

Parcels delivered per household in one week

Number of parcels	Number of households
1	24
2	14
3	9
4	6
5	4
6	3
Total	60

The frequency table shows the number of parcels delivered to each of 60 households during one week. Which of the following correctly compares the mean and the median of these 60 values?

(A) The mean is 0.35 less than the median.

(B) The mean is equal to the median.

(C) The mean is 0.35 greater than the median.

(D) The mean is 1.35 greater than the median.

105 Mark for Review

A drum of paint contains 180 kg of resin, which is 60% of the drum's total mass. Of the rest, 35% is filler and the remainder is pigment. How many kilograms of pigment does the drum contain?

ANSWER

106 Mark for Review

For a certain rectangular banner, the ratio of its height to its base is 9 : 6. If the base of the banner increases by 12 inches, how must the height change to maintain this ratio?

A It must decrease by 12 inches.

B It must increase by 12 inches.

C It must increase by 18 inches.

D It must decrease by 18 inches.

107 Mark for Review

Survey Question Results

Response	Number of members
Strongly agree	11
Agree	23
Neither agree nor disagree	14
Disagree	8
Strongly disagree	4
Total	60

For one week, the manager of an indoor climbing gym gathered member feedback by randomly selecting 60 members and asking them to complete a survey question. A total of 780 members used the gym during the week in which the survey was completed. The survey question and the survey results are shown below.

To what extent do you agree or disagree with the following statement: "I would recommend this climbing gym to others." (Select only one of the five choices.)

Which of the following is an appropriate conclusion about the members surveyed?

A. The majority of the members would recommend the climbing gym to others.

B. The majority of the members would not recommend the climbing gym to others.

C. The majority of the members would neither recommend nor not recommend the climbing gym to others.

D. The number of members who would recommend the climbing gym to others is the same as the number of members who would not recommend the climbing gym to others.

108 Mark for Review

A bakery tracked the number of loaves, rolls, and pastries it sold in a month. The number of rolls sold was 84% of the number of pastries sold, and the number of loaves sold was 25% of the number of rolls sold. If 4,200 loaves were sold, what was the sum of the numbers of rolls and pastries sold?

ANSWER

109 Mark for Review

During the first 6.00 seconds after a cargo sled started moving, its speed increased to 8.00 feet per second. From 6.00 seconds to 15.0 seconds after the sled started moving, its speed increased from 8.00 feet per second to 26.0 feet per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the sled, in feet per second per second, during the first 6.00 seconds after the sled started moving and the average rate of change of the speed of the sled, in feet per second per second, from 6.00 seconds to 15.0 seconds after the sled started moving?

ANSWER

110 Mark for Review

The lengths, in centimeters, of 19 salvaged copper rods at a recycling yard were recorded, and the mean, median, range, and standard deviation of the recorded lengths were found. The rod with the greatest recorded length was later found to actually be 8 centimeters longer than its recorded length. Which of these four values remains unchanged when the corrected length is used?

A Mean

B Median

C Range

D Standard deviation

111 Mark for Review

An equilateral triangle has side length 12 cm. If the side length is increased by 24%, by what percent does the perimeter increase?

ANSWER

112 Mark for Review

A researcher selected 2 random samples of members of a large cooperative to estimate the percentage of members who planned to vote in favour of building a new warehouse. Based on the first sample, the researcher estimated that 62% of the members would vote in favour, with an associated margin of error of 8.4%. Based on the second sample, the researcher estimated that 77% of the members would vote in favour, with an associated margin of error of 5.1%. Assuming the margins of error were calculated in the same way, which of the following best explains why the second sample obtained a smaller margin of error than the first sample?

A The first sample contained fewer members than the second sample.

B The first sample contained more members than the second sample.

C The first sample contained a lower percentage of members who planned to vote in favour of the new warehouse.

D The first sample contained a higher percentage of members who planned to vote in favour of the new warehouse.

113 Mark for Review

An 18-foot length of copper tubing and a 6-foot length of copper tubing were each cut completely into 9-inch pieces. How many more 9-inch pieces resulted from the 18-foot length than from the 6-foot length? (12 inches = 1 foot)

A 8

B 12

C 16

D 24

114 Mark for Review

A shipping port's land area is 55% container yards and 45% docks. Paved access lanes cover 28% of the container yard land and 12% of the dock land. What percent of the entire port's land area is paved access lanes?

ANSWER

115 Mark for Review

A biologist recorded the trunk diameter, x, in centimetres and the daily sap yield, y, in litres for a sample of maple trees. The line of best fit for these data is $y = 0.42x + 1.10$. The biologist then measured an additional tree with a trunk diameter of 6.0 cm and recorded a daily sap yield of 4.30 litres. Which of the following statements best describes this tree's recorded sap yield?

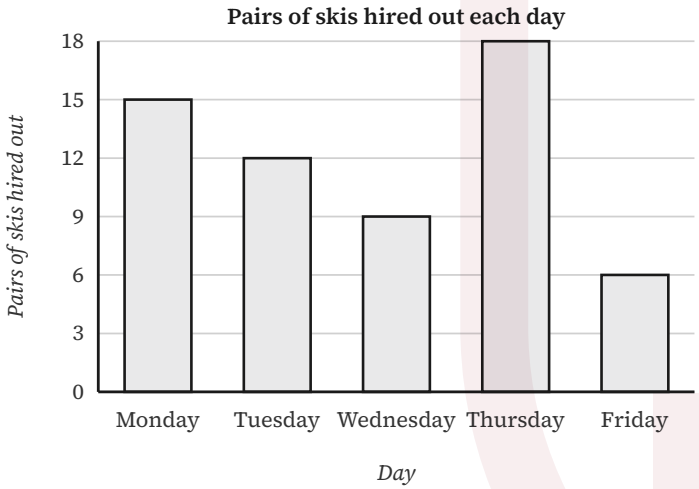
A. It is approximately 0.68 litres greater than the value predicted by the line of best fit.

B. It is approximately 0.68 litres less than the value predicted by the line of best fit.

C. It is approximately 1.10 litres greater than the value predicted by the line of best fit.

D. It is the value predicted by the line of best fit.

116 Mark for Review



The bar chart shows the number of pairs of skis a rental shop hired out on each day of one week, from Monday through Friday. The number of pairs hired out on Tuesday was what percent of the total number of pairs hired out during these five days?

A 10%

B 15%

C 20%

D 25%

117 Mark for Review

After a 25% increase, the monthly fee for a storage unit is 85 dollars. What was the monthly fee before the increase?

A $60.00

B $63.75

C $68.00

D $106.25

118 Mark for Review

A delivery van starts a route with 25 gallons of fuel in its tank. The van traveled at an average speed of 60 miles per hour for 3 hours and consumed fuel at a rate of 24 miles per gallon. What percent of the fuel in the tank was used for the 3-hour route?

A 28.0

B 30.0

C 31.5

D 32.5

119 Mark for Review

In a music library, 40% of the tracks are studio recordings and 60% are live recordings. Of the studio recordings 25% are instrumental, and of the live recordings 10% are instrumental. What percent of all the tracks in the library are instrumental?

ANSWER

120 Mark for Review

Poll results

Marlow	375
Denbury	265

The table shows the results of a poll. A total of 640 registered voters selected at random were asked which candidate they would vote for in an upcoming town council election. According to the poll, if 8,320 people vote in the election, by how many votes would Marlow be expected to win?

A 110

B 1,430

C 4,875

D 7,680

121 Mark for Review

For a study, a research team measured the shell lengths of 240 freshwater mussels from Lake Verrin and 120 freshwater mussels from Lake Osteen. The team determined that the mean shell length of the 240 mussels from Lake Verrin was 54 millimeters and the mean shell length of the 120 mussels from Lake Osteen was 63 millimeters. What was the mean shell length, in millimeters, of all 360 mussels the team measured for this study?

A 55

B 57

C 58.5

D 60

122 Mark for Review

A conservator is recording the dimensions of a stone staircase using an ancient unit called the cubit. If the staircase rises 34 cubits from its base to its top landing, what is that rise in inches? (1 cubit = 18 inches)

ANSWER

123 Mark for Review

A food bank warehouse holds 90 tonnes of stock, including 45 tonnes of grain and 20 tonnes of canned goods. Over the following month a donation drive increases the grain held by 16%, while distributions reduce the canned goods held by 15%. After that month, what is the combined mass, in tonnes, of grain and canned goods held?

ANSWER

124 Mark for Review

Preferred outing by response type

Preferred outing	Mailed card	Web form
Concert	50%	70%
Museum tour	20%	15%
Picnic	20%	10%
Boat ride	10%	5%
Total	100%	100%

A gardening club surveyed each of its members, asking every member to select the one outing he or she preferred for the club's next event. Some members responded on a mailed card, and the others responded on a web form. The table shows the distribution of preferred outing, in percent, for each response type used. For the survey, the number of web responses was three times the number of mailed responses. If a member who preferred a picnic is selected at random, what is the probability that the member responded on the web form?

ANSWER

125 Mark for Review

A researcher surveys 420 shoppers as they leave a single organic grocery store and finds that 75% of them cook dinner at home most nights. The researcher concludes that 75% of the adults in the town cook dinner at home most nights. Which of the following is the most appropriate conclusion from this research?

A. The finding applies only to shoppers at that grocery store, not to all adults in the town.

B. 75% of adults in the town cook dinner at home most nights, because the sample size is large.

C. The sample is unbiased because it includes 420 people.

D. The finding can be generalised to the town because the shoppers were surveyed on several different days.

126 Mark for Review

A town's annual budget allots 65% of its money to operations. Of the money not allotted to operations, 30% goes to debt service and the rest goes to capital projects. If debt service receives 42 million dollars, how many million dollars go to capital projects?

ANSWER

127 Mark for Review

A walkway bed is in the shape of a right rectangular prism. Its base has a width of 4 feet and a length of 18 feet, and it will be filled 18 inches deep with crushed gravel. The total cost of the gravel needed to fill the bed to this depth is 260 dollars. What is the unit cost, in dollars, per cubic yard, for the gravel? (1 yard = 3 feet; 1 foot = 12 inches)

ANSWER

128 Mark for Review

A store sells birdseed in two brands. Brand A is sold in 2.4-kilogram bags costing 7.20 dollars each, and Brand B is sold in 4-kilogram bags costing 11.20 dollars each. A customer wants to buy exactly 12 kilograms of birdseed and will buy whole bags of a single brand. How much money, in dollars, would the customer save by buying the brand with the lower price per kilogram rather than the other brand?

A 0.20

B 2.40

C 4.00

D 33.60

129 Mark for Review

Two volunteers are each conducting a survey among the 1,450 residents of a housing estate to determine the proportion of residents who would be interested in a new community garden. The first volunteer randomly selected 120 residents to survey, and the second volunteer randomly selected 60 residents to survey. In both surveys, 28% of residents indicated that they are interested. If F represents the margin of error for the first volunteer's sample proportion, S represents the margin of error for the second volunteer's sample proportion, and both volunteers have the same amount of confidence in their results, which of the following is most likely true?

A $F < S$

B $F > S$

C $F = S$

D The relationship between F and S cannot be determined.

130 Mark for Review

Fifteen largest hydroelectric stations by province and generating capacity

Province	Low	Medium	High	Total
Alta	3	1	2	6
Borea	1	1	0	2
Cendra	0	2	0	2
Dorna	1	0	1	2
Elmar	1	0	0	1
Fenwick	1	1	0	2
Total	7	5	3	15

The table shows the distribution, by province and generating capacity (maximum rate of power generation), of the fifteen largest hydroelectric stations in a certain country in 2020. The total generating capacity of the six stations located in Alta was 3,150 megawatts (MW), and the total generating capacity of the fifteen stations was 7,400 MW in 2020. The amount of energy produced in one hour at a rate of one megawatt is one megawatt-hour. If each of the six Alta stations in 2020 had operated continuously for 12 hours at its maximum rate of power generation, approximately how many megawatt-hours of energy would the six stations have produced?

A 300

B 3,000

C 7,000

D 40,000

131 Mark for Review

Data set P consists of 41 integers between 320 and 480. Data set Q consists of all the integers in data set P as well as the integer 15. Which of the following must be less for data set P than for data set Q?

I. The mean

II. The median

A I only

B II only

C I and II

D Neither I nor II

132 Mark for Review

A recycling load contains 180 kg of glass, which is 45% of the load's total mass. Of the remainder, 60% is paper and the rest is metal. How many kilograms of paper are in the load?

ANSWER

133 Mark for Review

A botanical garden wants to estimate the mean length of time its visitors spend in the garden. A researcher selected 200 of the garden's visitors at random and recorded the length of time each of them spent in the garden. The mean time for the selected visitors was 42.5 minutes, with an associated margin of error of 1.5 minutes. Which of the following conclusions is best supported by these results?

A. It is plausible that the mean time spent in the garden by all of the garden's visitors is between 41.0 minutes and 44.0 minutes.

B. It is plausible that the mean time spent in the garden by all of the garden's visitors is between 37.5 minutes and 40.5 minutes.

C. The mean time spent in the garden by the 200 selected visitors was between 44.0 minutes and 47.0 minutes.

D. The margin of error of 1.5 minutes indicates that the times recorded for the 200 selected visitors differed from one another by at most 1.5 minutes.

134 Mark for Review

A wildlife survey counted frogs, toads, and newts in a marsh. The number of toads counted was 75% of the number of newts counted, and the number of frogs counted was 30% of the number of toads counted. If 5,400 frogs were counted, what was the sum of the numbers of toads and newts counted?

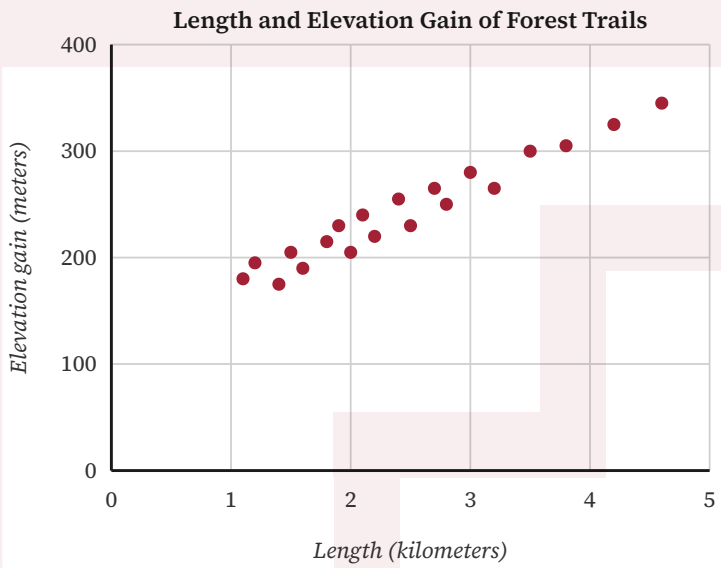
ANSWER

135 Mark for Review

A humidifier tank holds a solution that is 80% water and 20% mineral additive by volume. If 25% of the water evaporates, what percent of the remaining solution is water?

ANSWER

136 Mark for Review



The scatterplot shows the length x, in kilometers, and the total elevation gain y, in meters, for each of 20 hiking trails in a national forest. Which of the following could be an equation for a line of best fit for the data?

A $y = 20x + 120$

B $y = 50x$

C $y = 50x + 120$

D $y = 90x + 120$

137 Mark for Review

A regional power grid supplied 250 gigawatt-hours of electricity last month. Of that total, 60% came from natural gas, 16% came from wind, and the rest came from hydropower. This month the grid's natural gas output falls by 8% and its wind output rises by 30%. How many more gigawatt-hours of natural gas than wind does the grid supply this month?

ANSWER

138 Mark for Review

Residence and most frequent mode of travel to campus

Residence	Bus	Bicycle	Car	Total
Lives on campus	40	120	20	180
Lives off campus	150	40	230	420
Total	190	160	250	600

The table shows the 600 students at a college classified by where they live and by the mode of travel each uses most often to get to campus. One of these 600 students is selected at random. Given that the selected student travels to campus most often by bicycle, what is the probability that the student lives on campus?

A $\frac{1}{5}$

B $\frac{4}{15}$

C $\frac{2}{3}$

D $\frac{3}{4}$

139 Mark for Review

A rectangular deck measures 6.0 meters by 2.5 meters. If 1 liter of sealant covers approximately 30,000 square centimeters, how many liters of sealant are needed to coat the deck twice? (1 meter = 100 centimeters)

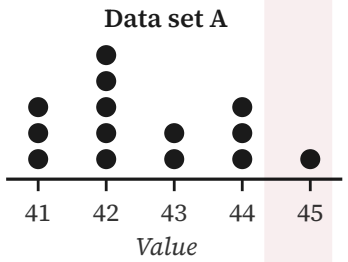
ANSWER

140 Mark for Review

A solar farm generated 480 megawatt-hours in June, which was 60% of its output for the whole quarter. Of the remaining output for the quarter, 40% was generated in July and the rest was generated in August. How many megawatt-hours did the solar farm generate in August?

ANSWER

141 Mark for Review



The dot plot represents the 14 values in data set A. Data set B is created by adding 38 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

142 Mark for Review

The director of a museum selected a sample of 156 visitors at random to estimate the percentage of the museum's visitors who would buy an annual pass if one were offered. From the survey, the director estimates that 74% of the museum's visitors would buy an annual pass, with an associated margin of error of 6.88%. If the survey is repeated with a random sample of 312 visitors and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

A The margin of error will be lower.

B The margin of error will be higher.

C The estimate of the percentage of the museum's visitors who would buy an annual pass will be lower.

D The estimate of the percentage of the museum's visitors who would buy an annual pass will be higher.

143 Mark for Review

A "slow" clock falls behind at the same rate every hour. It is set to the correct time at 6:00 AM. When the clock shows 7:00 AM the same day, the correct time is 7:12 AM. When the clock shows 11:30 AM that day, what is the correct time?

A 11:42 AM

B 12:06 PM

C 12:30 PM

D 12:36 PM

144 Mark for Review

A streaming service's catalog contains 2,000 hours of programming. Of this total, 60% is films, 20% is documentaries, and the rest is series. Over the next year the number of hours of films increases by 15% and the number of hours of documentaries decreases by 25%. After these changes, how many more hours of films than documentaries are in the catalog?

ANSWER

Answer key

Letters are multiple choice; numbers are student produced responses. Where two forms are listed, either is accepted on test day.

Q	ANSWER
1	117.3
2	C
3	A
4	C
5	C
6	D
7	D
8	3200
9	C
10	60
11	6000
12	C
13	B
14	B
15	91.9
16	A
17	48.8

Q	ANSWER
18	126
19	B
20	32500
21	C
22	144
23	D
24	A
25	A
26	120
27	C
28	B
29	24
30	B
31	19
32	C
33	C
34	61

Q	ANSWER
35	B
36	C
37	80
38	D
39	100
40	18.8
41	D
42	C
43	C
44	54.5
45	A
46	13.6
47	D
48	80
49	B
50	A
51	27

Q	ANSWER
52	28
53	D
54	A
55	235.6
56	B
57	C
58	27000
59	D
60	179
61	B
62	D
63	B
64	13.4
65	A
66	9.2
67	4
68	D

Q	ANSWER
69	114
70	4.5
71	2.4
72	B

Q	ANSWER
73	A
74	D
75	12
76	D
77	D
78	C
79	B
80	25
81	B
82	D
83	A
84	90
85	40
86	C
87	B
88	C
89	95680

Q	ANSWER
90	44
91	B
92	B
93	27
94	C
95	A
96	69
97	600
98	D
99	25
100	D
101	40
102	204
103	B
104	C
105	78
106	C

Q	ANSWER
107	A
108	36800
109	0.67
110	B
111	24
112	A
113	C
114	20.8
115	A
116	C
117	C
118	B
119	16
120	B
121	B
122	612
123	69.2

Q	ANSWER
124	3/5
125	A
126	98
127	65
128	B
129	A
130	D
131	D
132	132
133	A
134	42000
135	75
136	C
137	86
138	D
139	10
140	192

Q	ANSWER
141	C
142	A
143	D
144	1080

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